QP CODE **Reg.No-:-.....**
Name-:-.....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) REGULAR EXAMINATION MARCH 2026

Fourth Semester

Discipline Specific Elective Course
MG4DSEBOTA08 Artificial Intelligence in Plant Science

(2024 ADMISSION ONWARDS)

Duration: 2 Hours- Maximum Marks: 70

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

[Learning Domain][CO No(s)]

Part A

Short Answer

Answer any 10 questions

Each question carries **2** marks

1. Define Artificial Intelligence. [K][CO1]
2. Name any two AI learning models. [K][CO1]
3. What is 'data' in AI? [K][CO1]
4. Mention one use of cloud computing in AI. [K][CO1]
5. State one relevance of AI to plant taxonomy. [K][CO1]
6. What is image analysis in plant identification? [U][CO2]
7. Name any one CNN-based plant ID app. [U][CO2]
8. What is crop forecasting in AI? [U][CO2]
9. Name one AI quality standard body in agriculture. [U][CO2]
10. Mention one disease detected by AI in crops. [U][CO3]
11. Name any one mobile app used by farmers for plant disease detection. [A][CO4]
12. Mention two ways AI can support environmental sustainability in agriculture. [A][CO4]

(10 × 2 = 20)

Part B

Short Essay Type Questions

Answer 6 questions.

Each question carries 5 marks

13. Explain the scope of Artificial Intelligence in science. [U][CO1]
14. Explain the role of AI in crop classification and stress identification. [U][CO1]
15. Describe the key components of AI with examples. [U][CO1]

16. Show how CNNs are used in plant identification. [A][CO2]
17. Explain how AI integrates weather and soil data for crop forecasting. [E][CO3]
18. Outline the role of AI + IoT in irrigation planning. [An][CO3]
19. Explain how AI supports climate resilience in agriculture. [E][CO3]
20. Write the importance of agri-tech startups using AI for sustainable farming. [A][CO3]
21. Write the benefits and limitations of mobile plant disease apps. [A][CO4]
22. Explain ethical issues in AI-based plant disease prediction. [U][CO4]

(8 × 5 = 40)

Part C

Essay Type Questions

Answer 1 questions.

The question carries 10 marks

23. Explain the major AI learning models with examples from plant science. [U][CO2]
24. Evaluate the role of AI in crop forecasting, disease detection, and sustainable agriculture with real-world examples. [E][CO3]

(1 × 10 = 10)

END OF THE QUESTION PAPER

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